SmartTask : Context-Aware Mobile To-Do List for Intelligent Task Management

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SmartTask

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SmartTask : A Context-Aware Mobile Task Management System

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Problem

Solution

How the Solution Addresses the Problem?

Most to-do apps ignore user context, causing poorly timed reminders, interruptions, and extra mental load.

Solution :

- **Smart Prioritization →** Tasks are automatically prioritized using real-time user context
- **Smart Notification Timing →** Notifications are delivered only when the user is receptive
- **Focus-Aware Alerts →** Alerts are suppressed during meetings, travel, or focus time
- **Reduced Cognitive Load →** System shows only the most relevant tasks

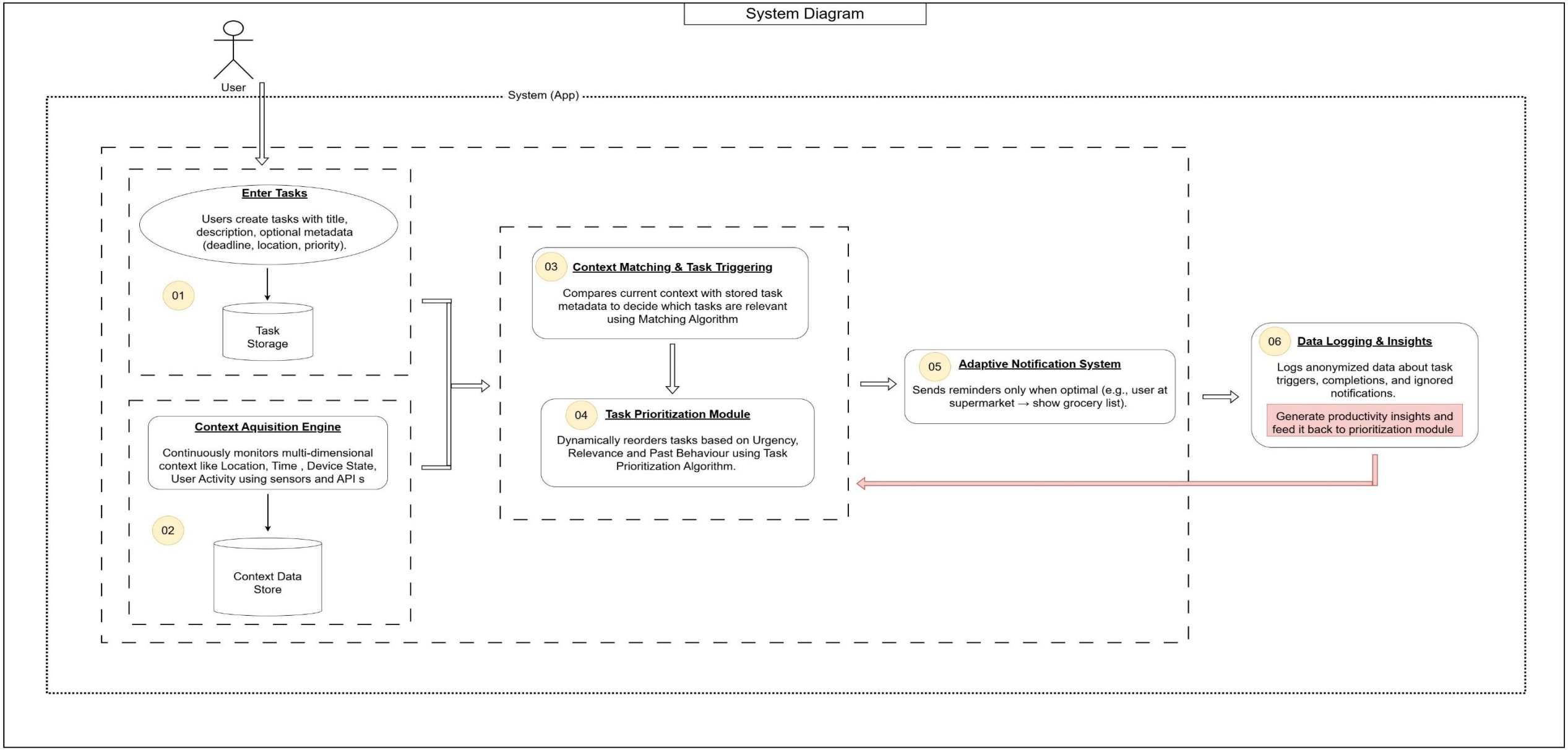
Solution Overview

SmartTask is a context-aware mobile to-do app that prioritizes tasks and sends reminders based on the user's real-time context such as location, activity, and schedule.

Key Features :

- **Context-Aware Task Prioritization** → Tasks are automatically ranked using user context
- **Adaptive Notification System**→ Reminders are sent at the most appropriate moments to avoid interruptions.
- **Behavioral Learning & Insights** → The system learns from user actions to improve future task suggestions. (future improvement)

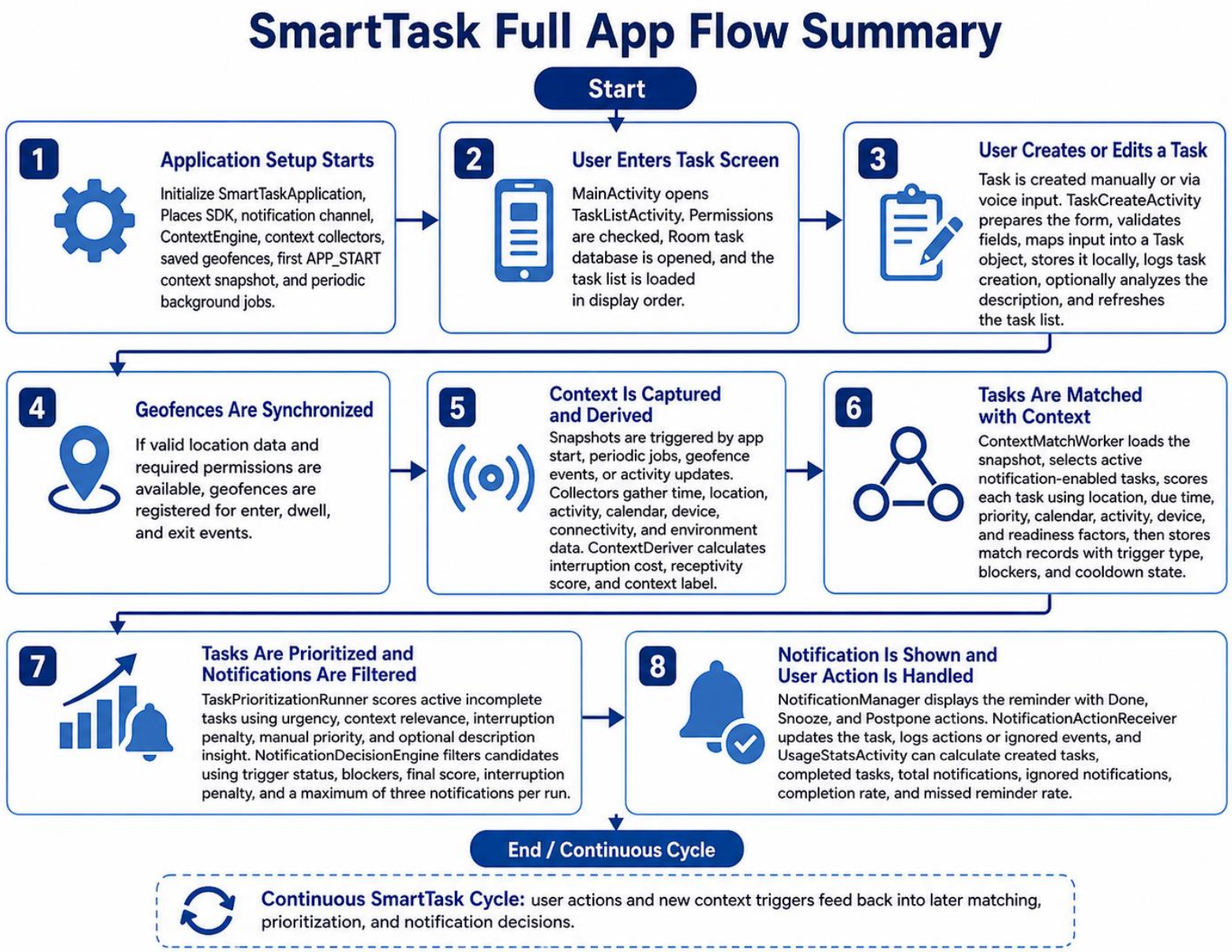
System Design: Six - Component Architecture



System Design: Components

- **Task Input –** manual task creation and editing
- **Voice Command Task Creation –** captures spoken task commands and converts them into structured task fields. (additional component suggested by examiner panel)
- **Context Acquisition Engine –** gather current user context
- **Context Matching & Triggering –** filter tasks relevant to the current context
- **Task Prioritization–** ranks tasks based on urgency and relevance
- **Adaptive Notification System –** sends reminders at suitable times
- **Data Logging & Insights –** records user actions for future improvement

System Design : App Flow In Real World



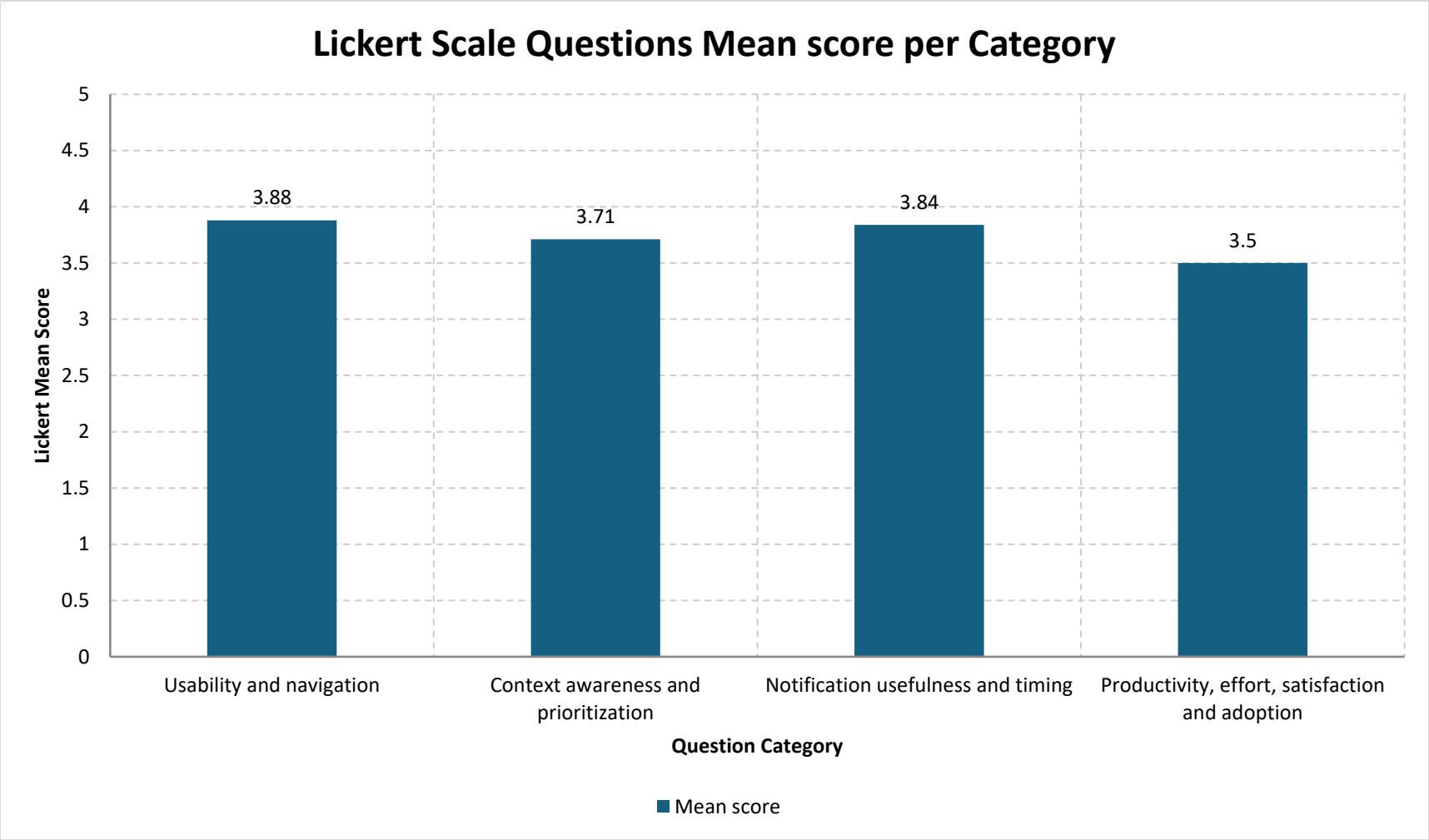
PP2 Comments and Actions Taken

Comment	Action Taken
Consider task description when calculating prioritization scores	• Task description and title is analyzed when a task is created or edited and then scores are assigned accordingly. • That score is used when final task priority score is calculated.
Provide testimonials from users	• A user evaluation was conducted. (To View Results Analysis, Click Here.)

User Evaluation Results

Metric	Value
Participants	8
Total active testing days	109
Average active days per participant	13.62
Active days range	12 to 15
Total tasks created	211
Average tasks created per participant	26.38
Total tasks completed	185
Average tasks completed per participant	23.12
Overall task completion rate	87.7%
Mean participant completion rate	88.6%
Completion rate range	81.6% to 96.2%
Total notifications received	194
Average notifications received per participant	24.25
Total notifications ignored or missed	32
Average ignored or missed notifications per participant	4
Overall ignored or missed notification rate	16.5%
Mean participant ignored or missed rate	16.2%
Ignored or missed rate range	9.1% to 28.1%
Overall Likert mean	3.73

User Evaluation Results



Limitations and Future Work

Limitations	Future Work
Evaluation was done with only 8 valid participants.	Test with a larger and more diverse user group.
Testing period was limited to around two weeks.	Conduct a longer study to measure long term productivity impact.
No control group was used for comparison.	Compare SmartTask with traditional to do list apps.
Context accuracy depends on sensors, permissions, and location services.	Improve fallback handling for denied permissions and inaccurate sensor data.
Full behavioral learning was not implemented.	Add a stronger learning model based on long term user patterns.
Data Logging and Insights was not implemented as a full user facing feature.	Add insights such as completion trends, ignored reminders, and productive time periods.
Background context checks can affect battery and resource usage.	Optimize background processing for better long term daily use.

Commercialization Potential

Business Model – How We Make Money

- **Differentiation:** Unlike other apps, ***SmartTask*** is adaptive & context-aware, offering intelligent task prioritization and notifications tuned to real-life conditions.
- **Investment & Cost:**
  - Labor: Development hours for AI algorithms, UI/UX, backend.
  - Material: Android devices for testing.
  - API s : Subscriptions Cost for API s (e.g. google maps).
  - Recovery: Freemium approach – free version with ads; premium subscription offers advanced insights and an ad-free experience.
- **Target Market:** Students, busy professionals, productivity enthusiasts.
- **Customer Profile:** Mobile-first users with heavy task-list dependency and fast-paced lifestyles.
- **Value Proposition:** Reduce stress, improve productivity, and deliver notifications at the right moment, not just the next moment.
- **IP Considerations:** Proprietary algorithms for context-matching and adaptive notification timing create defensible intellectual property.

App Demo

Q & A ??

Thank You !